Assignment
Subject: programing_fundamental_lab_3
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Lab Objective: Expressions, type casting, coercion, formatting, random numbers.

Task1:

Assume that the following variables are defined: `int age; double pay; char section;` Write a single cin statement that will read input into each of these variables.

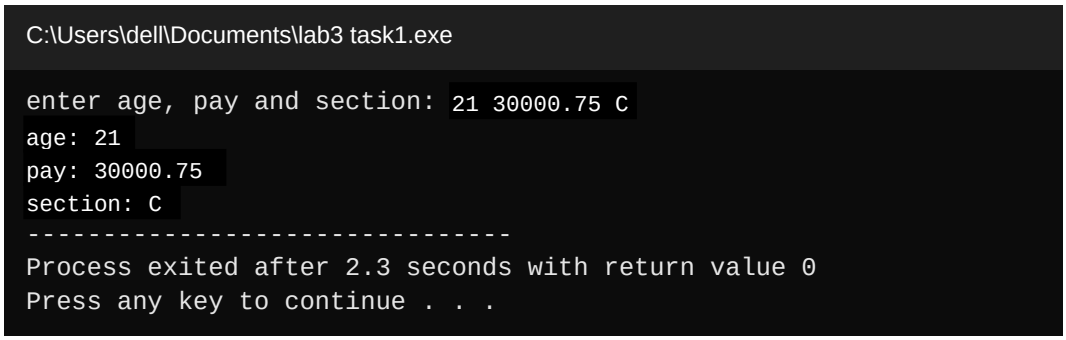
Single cin statement:

```
cin >> age >> pay >> section;
```

Complete program:

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
    int age;
    double pay;
    char section;
    cout<<"enter age, pay and section: ";
    cin>>age>>pay>>section;
    cout<<"age: "<<age<<endl;
    cout<<fixed<<setprecision(2);
    cout<<"pay: "<<pay<<endl;
    cout<<"section: "<<section<<endl;
    return 0;
}
```

Output:



Task2:

Complete the following table by writing the value of each expression in the Value column according to C++ language rules.

Expression	Value
28 / 4 - 2	5
6 + 12 * 2 - 8	22
4 + 8 * 2	20
6 + 17 % 3 - 2	6
2 + 22 * (9 - 7)	46
(8 + 7) * 2	30
(16 + 7) % 2 - 1	0
12 / (10 - 6)	3
(19 - 3) * (2 + 2) / 4	16

Program used to verify the values:

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
    cout<<left<<setw(26)<<"28 / 4 - 2"<<"= "<<(28 / 4 - 2)<<endl;
    cout<<left<<setw(26)<<"6 + 12 * 2 - 8"<<"= "<<(6 + 12 * 2 - 8)<<endl;
    cout<<left<<setw(26)<<"4 + 8 * 2"<<"= "<<(4 + 8 * 2)<<endl;
    cout<<left<<setw(26)<<"6 + 17 % 3 - 2"<<"= "<<(6 + 17 % 3 - 2)<<endl;
    cout<<left<<setw(26)<<"2 + 22 * (9 - 7)"<<"= "<<(2 + 22 * (9 - 7))<<endl;
    cout<<left<<setw(26)<<"(8 + 7) * 2"<<"= "<<((8 + 7) * 2)<<endl;
    cout<<left<<setw(26)<<"(16 + 7) % 2 - 1"<<"= "<<((16 + 7) % 2 - 1)<<endl;
    cout<<left<<setw(26)<<"12 / (10 - 6)"<<"= "<<(12 / (10 - 6))<<endl;
    cout<<left<<setw(26)<<"(19 - 3) * (2 + 2) / 4"<<"= "<<((19 - 3) * (2 + 2) / 4)<<endl;
    return 0;
}
```

Output:

```
C:\Users\dell\Documents\lab3 task2.exe

28 / 4 - 2                 = 5
6 + 12 * 2 - 8             = 22
4 + 8 * 2                  = 20
6 + 17 % 3 - 2             = 6
2 + 22 * (9 - 7)           = 46
(8 + 7) * 2                = 30
(16 + 7) % 2 - 1           = 0
12 / (10 - 6)              = 3
(19 - 3) * (2 + 2) / 4     = 16
--------------------------------
Process exited after 2.3 seconds with return value 0
Press any key to continue . . .
```

Task3:

Assume a program has the variables `int units; float mass; double weight;` and the statement `weight = mass * units;` Which automatic data type conversion will take place?

A. mass is demoted to an int, units remains an int, and the result of mass * units is an int.
B. units is promoted to a float, mass remains a float, and the result of mass * units is a float.
C. units is promoted to a float, mass remains a float, and the result of mass * units is a double.

Answer: B. units (int) is promoted to float, so mass * units is a float (4 bytes). Only after that is the result converted to double when it is assigned to weight.

```
#include<iostream>
using namespace std;
int main()
{
    int units = 6;
    float mass = 2.5;
    double weight;
    weight = mass * units;
    cout<<"size of (mass * units) in bytes: "<<sizeof(mass * units)<<endl;
    cout<<"size of weight in bytes: "<<sizeof(weight)<<endl;
    cout<<"weight = "<<weight<<endl;
    return 0;
}
```

Output:

```
C:\Users\dell\Documents\lab3 task3.exe

size of (mass * units) in bytes: 4
size of weight in bytes: 8
weight = 15
--------------------------------
Process exited after 2.3 seconds with return value 0
Press any key to continue . . .
```

Task4:

Assume `int a, b = 2; float c = 4.2;` and the statement `a = b * c;` What value will be stored in a? A. 8.4 B. 8 C. 0 D. None of the above

Answer: B (8). b * c is 8.4, but a is an int, so the fractional part is dropped when it is assigned.

```
#include<iostream>
using namespace std;
int main()
{
    int a, b = 2;
    float c = 4.2;
    a = b * c;
    cout<<"b * c = "<<b * c<<endl;
    cout<<"value stored in a = "<<a<<endl;
    return 0;
}
```

Output:

```
C:\Users\dell\Documents\lab3 task4.exe

b * c = 8.4
value stored in a = 8
--------------------------------
Process exited after 2.3 seconds with return value 0
Press any key to continue . . .
```

Task5:

Assume that qty and salesReps are both integers. Use a type cast expression to rewrite `unitsEach = qty / salesReps;` so it will no longer perform integer division.

```
unitsEach = static_cast<double>(qty) / salesReps;
```

```
#include<iostream>
using namespace std;
int main()
{
    int qty, salesReps;
    double unitsEach;
    cout<<"enter qty and salesReps: ";
    cin>>qty>>salesReps;
    cout<<"integer division: "<<qty / salesReps<<endl;
    unitsEach = static_cast<double>(qty) / salesReps;
    cout<<"with type cast: "<<unitsEach<<endl;
    return 0;
}
```

Output:

```
C:\Users\dell\Documents\lab3 task5.exe

enter qty and salesReps: 53 4
integer division: 13
with type cast: 13.25
--------------------------------
Process exited after 2.3 seconds with return value 0
Press any key to continue . . .
```

Task6:

Math Tutor: Write a program that displays two random numbers to be added, pauses while the student works on the problem, and then displays the correct solution when a key is pressed.

```
#include<iostream>
#include<cstdlib>
#include<ctime>
using namespace std;
int main()
{
    srand(time(0));
    int a = rand() % 900 + 100;
    int b = rand() % 900 + 100;
    cout<<"   "<<a<<endl;
    cout<<" + "<<b<<endl;
    cout<<"  ------"<<endl;
    cout<<"press enter to see the answer...";
    cin.get();
    cout<<endl;
    cout<<"   "<<a<<endl;
    cout<<" + "<<b<<endl;
    cout<<"  ------"<<endl;
    cout<<"   "<<a + b<<endl;
    return 0;
}
```

Output:

```
C:\Users\dell\Documents\lab3 task6.exe

   623
 + 357
  ------
press enter to see the answer...
   623
 + 357
  ------
   980

--------------------------------
Process exited after 2.3 seconds with return value 0
Press any key to continue . . .
```

Home Tasks

Task 1: Each of the following programs has some errors. Locate as many as you can.

Program-1 errors:

1. The line `#include <iostream>` is missing.
2. `void main()` should be `int main()`.
3. `Cin` should be `cin` (C++ is case sensitive), and the operator should be >> not << for input.
4. `cin << number2;` must be `cin >> number2;`
5. `number1 + number2 = sum;` must be `sum = number1 + number2;`
6. The last cout is missing << after cout and has no semicolon at the end.

Corrected Program-1:

```
#include<iostream>
using namespace std;
int main()
{
    double number1, number2, sum;
    cout << "Enter a number: ";
    cin >> number1;
    cout << "Enter another number: ";
    cin >> number2;
    sum = number1 + number2;
    cout << "The sum of the two numbers is " << sum;
    return 0;
}
```

Output:

```
C:\Users\dell\Documents\lab3 home program1.exe

Enter a number: 12.5
Enter another number: 18.75
The sum of the two numbers is 31.25
--------------------------------
Process exited after 2.3 seconds with return value 0
Press any key to continue . . .
```

Program-2 errors:

1. `void main()` should be `int main()`.
2. `float<static_cast>(number1)` is written the wrong way round; it should be `static_cast<float>(number1)`.
3. The line `cout << quotient` has no semicolon.

Corrected Program-2:

```
#include <iostream>
using namespace std;
int main()
{
    int number1, number2;
    float quotient;
    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";
    cin >> number1 >> number2;
    quotient = static_cast<float>(number1) / number2;
    cout << quotient;
    return 0;
}
```

Output:

```
C:\Users\dell\Documents\lab3 home program2.exe

Enter two numbers and I will divide
the first by the second for you.
25 4
6.25
--------------------------------
Process exited after 2.3 seconds with return value 0
Press any key to continue . . .
```

Task2:

Average of Values: store the values 36, 41, 29, 38 and 31 in five different variables. Calculate their sum in a variable named sum, then divide sum by 5 to get the average and display it.

```
#include<iostream>
using namespace std;
int main()
{
    int v1 = 36, v2 = 41, v3 = 29, v4 = 38, v5 = 31;
    int sum;
    double average;
    sum = v1 + v2 + v3 + v4 + v5;
    average = sum / 5.0;
    cout<<"sum = "<<sum<<endl;
    cout<<"average = "<<average<<endl;
    return 0;
}
```

Output:

C:\Users\dell\Documents\lab3 home task2.exe

```
sum = 175
average = 35
--------------------------------
Process exited after 2.3 seconds with return value 0
Press any key to continue . . .
```